Spherical Cyclic Time v2.1-E
Public Patch After Transfer Route Decision

Internal technical addendum

2026-07-04

Abstract

This addendum updates the public transfer-kernel language of *Spherical Cyclic Time* after the v2.0 preregistered transfer tests and the v2.1 mechanism/branch decisions. The earlier gaussian/diffusive form, $T_R(k) = \exp[-(k/k_c)^2]$, is no longer treated as the default closed-background transfer expectation. The conservative mainline is now an unspecified, model-dependent transfer operator $T_R(k)$. The gaussian form remains allowed only as a conditional open-system or diffusive toy limit, while the plateau-like form identified in v2.1 is recorded only as a toy explanatory branch for the closed deterministic proxy route. No physical cosmology, observational prediction, entropy-map derivation, or general no-go theorem is claimed.

1 Reason for this patch

The external referee-style review of the v1.7–v1.9 branch correctly distinguished two kinds of negative result. The v1.7–v1.9 work is best described as a requirements and obstruction audit: it documents why a full physicalization through an independently specified background, environment, state, spectral density, and entropy map was blocked. It is not a computational negative result.

The v2.0 branch was therefore redirected to a minimal falsifiable target: test whether simple closed-background toy transitions produce a gaussian transfer shape. Two background families were preregistered and executed without post-hoc changes. Both returned non-gaussian, bounded, high-k plateau-like transfer shapes.

2 Evidence from the preregistered v2.0 tests

The two tested families were:

1. a smooth power bounce, $a(y) = (1 + y^2)^{\alpha/2}$;
2. a compact C^2 matched power-law bounce.

Both used the same proxy equation,

$$u_k'' + \left[k^2 - \frac{a''}{a}\right] u_k = 0, \tag{1}$$

with the same k grid, transfer metric, and classification protocol. The primary value was $\alpha = 0.20$, with preregistered sensitivities $\alpha = 0.10$ and $\alpha = 0.50$.

Family	α	Classification	High-k plateau
Smooth power bounce	0.20	Non-gaussian	0.496709
Compact C^2 matched bounce	0.20	Non-gaussian	0.484705

Table 1: Primary preregistered transfer tests. The sensitivities $\alpha = 0.10$ and $\alpha = 0.50$ were also classified as non-gaussian in both families.

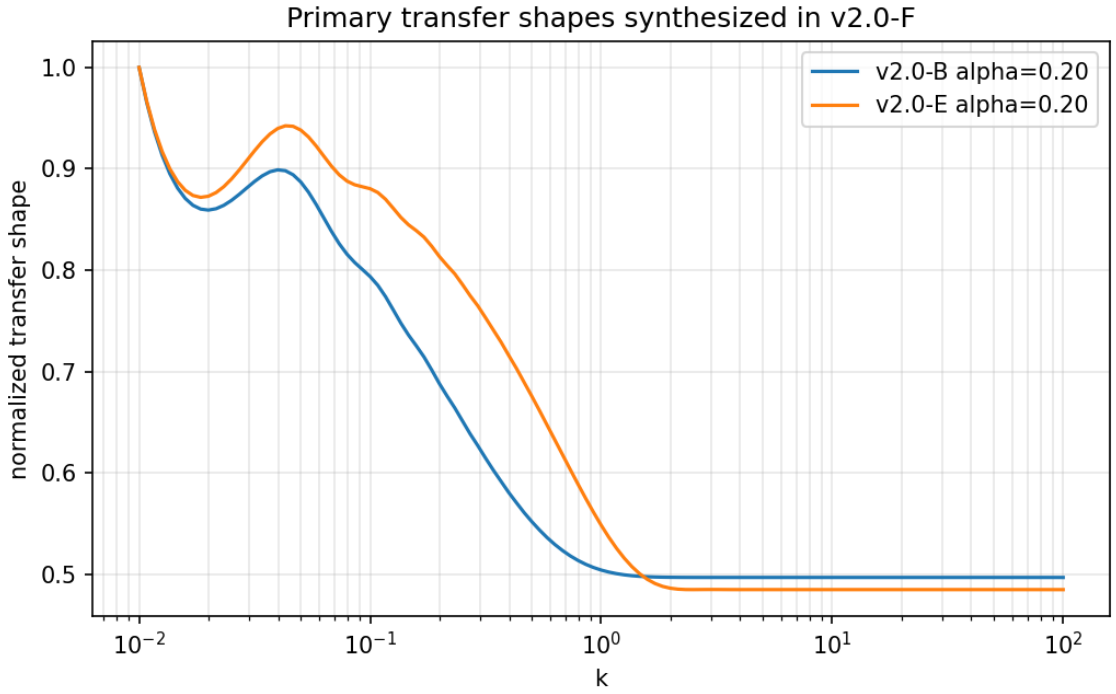


Figure 1: Primary transfer-shape comparison from v2.0-F. Both preregistered background families show bounded plateau-like behavior rather than gaussian damping.

3 Mechanism identified in v2.1-A

The plateau is not treated here as a new universal physical effect. For the closed deterministic proxy equation, set

$$V(y) = \frac{a''}{a}, \qquad E_k = u_k^2 + \left(\frac{u_k'}{k}\right)^2. \tag{2}$$

A direct differentiation gives

$$\frac{dE_k}{dy} = \frac{2V(y)u_k u_k'}{k^2}. \tag{3}$$

Thus, when $k^2 \gg |V(y)|$, the raw transfer is approximately transparent at high k, up to oscillatory and order-k^{-2} corrections. Under the v2.0 normalization, the high-k plateau follows from high-k

transparency divided by low-k reference amplification:

$$T_{\text{shape,high-k}} \simeq \frac{P_{\text{raw,high-k}}}{P_{\text{raw}}(k_{\min})} \simeq \frac{1}{P_{\text{raw}}(k_{\min})}. \tag{4}$$

This explains why a closed deterministic scattering-style proxy does not naturally produce irreversible gaussian damping.

4 Mainline transfer-language patch

The conservative mainline language is now:

$$T_R(k) = \text{model-dependent transfer operator.} \tag{5}$$

This replaces any presentation in which the gaussian form

$$T_R(k) = \exp[-(k/k_c)^2] \tag{6}$$

is treated as the default closed-background expectation.

The branch taxonomy is:

- **Closed deterministic background route:** the current evidence from two preregistered toy families points to bounded, plateau-like transfer, not gaussian damping.
- **Open-system/diffusive route:** the gaussian kernel remains a conditional toy limit only if an independent dissipative, diffusive, or coarse-graining mechanism is specified.
- **Plateau-like closed-route toy branch:** a minimal explanatory form is

  $$T_{\text{pl}}(k) = p_\infty + \frac{1 - p_\infty}{1 + (k/k_x)^2}, \tag{7}$$

  where p_∞ is the normalized high-k plateau. This is not a physical universal kernel.
- **More physical matching route:** execution remains blocked until a background/action, gauge-invariant variable, $z(\eta)$, matching surface, and matching conditions are independently specified.

5 Claim boundary

This patch does *not* claim a physical cyclic cosmology, an observational prediction, a no-go theorem for cyclic or bouncing backgrounds, a derived entropy map, or a universal transfer kernel. It records a methodological correction: the gaussian kernel is no longer the conservative default for the closed-background branch.

6 Public status

The appropriate public summary is: v1.7–v1.9 are requirements/survey obstruction audits; v2.0 is a limited negative computational result for two preregistered closed-background toy families; v2.1 explains the plateau mechanism and updates the transfer-kernel language accordingly.